Lab 03: Chaotic Text Generator

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Repository: https://github.com/toubani12/chaotic-text-generator
Application: https://<your-app>.streamlit.app

Problem and solution

A generated sentence has no reference answer, so a text generator cannot be judged from a few examples. We built a web test bench that samples hundreds of sentences from a part-of-speech HMM (plain, role-split or second-order states) or from a trigram model, shows the hidden tag path behind each sentence, and scores every batch with four metrics. A Viterbi tagger recovers the tags of any typed sentence. The same HMM is then tested on real English (UD English EWT) to see how far the synthetic results transfer.

Metrics at T = 1.0 (500 sentences, seed 42)

Model	Valid pattern	Plausible S-V	Novel	Distinct-2
HMM, plain states	16.2%	14.8%	99.8%	0.198
HMM, role-split states	100.0%	60.6%	99.4%	0.148
Trigram	57.0%	76.0%	91.2%	0.105
HMM, second order (extension)	20.0%	13.8%	99.8%	0.201

Plain HMM, 16.2%. A first-order model only knows the previous tag. After a NOUN it moves to VERB (0.39), PREP (0.22) or PUNCT (0.39) whatever the role of the noun, so a subject can end the sentence before the verb and an object can take a second verb.

Role-split HMM, 100.0%. NOUN_S is always followed by VERB, NOUN_O by PREP or PUNCT and NOUN_P by PUNCT: every path the model can sample is one of the 16 training patterns. The second-order HMM reaches only 20.0%, because the pair (DET, NOUN) is the same for a subject and an object.

Choice of temperature

- **Writing toy: role-split HMM at T = 1.5.** Structure stays at 100.0% while distinct-2 rises to 0.161 (from 0.148 at T = 1.0); surprising word pairs are acceptable in a toy.
- **Autocomplete: trigram at T = 0.5.** It is the only model that continues the user's words, and a low temperature gives the most predictable continuation (63.6% valid patterns, against 44.2% at T = 2.5).

Two generated sentences

> good “the teachers eat the birds near the river .” (role-split HMM, T = 1.0): valid pattern, subject-verb pair seen in training, not copied.

> bad: structure “the quiet school .” (plain HMM, T = 1.0): the tag path DET ADJ NOUN PUNCT is not a sentence. Cause: after the NOUN the model chose PUNCT (probability 0.39) without knowing that no verb had been produced yet.

Other failure types seen: meaning, a valid pattern with an odd pair (39.4% of role-split sentences); memorisation, “the children visit the garden .” copied by the trigram at T = 0.5.

What state splitting changed, and what it did not

Splitting NOUN, ADJ and DET by role gives the first-order model the memory it lacked: where it is in the sentence. Valid tag patterns rise from 16.2% to 100.0% and plausible subject-verb pairs from 14.8% to 60.6%, because subject nouns are now drawn only from subject positions. It did not add meaning: nouns and verbs are still drawn independently, so the trigram stays better on plausibility (76.0%); diversity drops (distinct-2 0.148 against 0.198); tagging accuracy was already 100%; and the states are written by hand for this grammar.

Synthetic corpus against real text (UD English EWT, 12,544 training and 2,077 test sentences)

Model	Accuracy	Known words	Unknown words	Whole sentence
HMM + Viterbi, synthetic corpus	100%	100%	no unknown words	100%
Most-frequent tag, EWT	86.2%	90.8%	28.6%	31.9%
HMM + Viterbi, EWT	88.0%	92.9%	28.2%	40.4%
HMM + suffix guessing, EWT (extension)	91.2%	93.3%	64.3%	49.0%

The synthetic grammar is unambiguous once the previous tag is known and every test word is in the 61-word vocabulary, so Viterbi is perfect. On real text the HMM gains only 1.9 points over the most-frequent-tag baseline. Unknown words (7.5% of tokens) are tagged correctly only 28.2% of the time: the smoothed emission k / (count + k·V) is largest for rare tags, so unknown proper nouns become X or INTJ. With 10,359 distinct tag sequences, the valid-pattern metric has no meaning on real text. Guessing unknown words from their suffix raises unknown-word accuracy to 64.3%. The guide lists 88.1% for the HMM; the current treebank download gives 88.0% with the unchanged code.

The corpus is synthetic and the grammar is tiny; metrics such as valid tag pattern are defined relative to this corpus and do not measure the quality of real English. All numbers are produced by the application and by src/evaluate.py and src/evaluate_real.py.